



SECTION 508 ACCESSIBILITY GUIDANCE
SUPPORTING INCLUSIVE ICT DELIVERY...

iCUBE Recommended Accessibility Testing and Validation Tools (2026)

RECOMMENDED VALIDATION APPROACH

- Automated Testing
- Manual Keyboard Testing
- Screen Reader Testing
- Color Contrast Verification
- Responsive Zoom Testing
- PDF Accessibility Validation



STANDARDS SUPPORTED

- Section 508
- WCAG 2.0 Level A/AA
- WCAG 2.1 (selected testing practices, where applicable)
- PDF/UA (where applicable)

ABOUT THIS GUIDE

- Developed and maintained by iCUBE, Inc.
- This guide reflects iCUBE's recommended accessibility testing practices. It is intended as a practical reference and is not an official U.S. Government publication.

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Repository: <https://github.com/icube-inc/accessibility-tools>

Purpose: This practical reference guide presents **iCUBE's recommended accessibility testing and validation tools** to help teams identify accessibility issues, evaluate conformance, and support inclusive digital experiences. The guide covers websites, web applications, documents, and **Information and Communication Technology (ICT)** solutions, with practices aligned to **Section 508** and **WCAG** requirements.

The guide is based on **Section 508 requirements and WCAG 2.0 Level A/AA success criteria**, while incorporating testing practices applicable to **WCAG 2.1**. iCUBE recommends a comprehensive validation approach combining **automated testing, manual inspection, keyboard-only testing, color contrast analysis, and assistive technology testing**.

Where formal accessibility documentation is required, iCUBE recommends using the **ITI Voluntary Product Accessibility Template (VPAT®)** to document testing results and supporting evidence and produce an **Accessibility Conformance Report (ACR)** aligned with federal procurement practices.

A **VPAT® is not a testing tool**. It is a reporting template developed by ITI to document ICT accessibility conformance. A completed VPAT® becomes an **Accessibility Conformance Report (ACR)**.

Note: This guide and tool mapping were developed by **iCUBE, Inc.** as an original compilation of publicly available Section 508/WCAG guidance, federal resources, and vendor documentation. The organization, categorization, and recommendations were prepared specifically for this guide. Accessibility standards, tool names, and vendor materials remain the property of their respective owners.



iCUBE Recommended Accessibility Testing and Validation Tools (2026)

Note: Tool inclusion reflects iCUBE’s recommendations and experience with accessibility testing practices. Inclusion does not represent endorsement, certification, or affiliation with any listed tool vendor.

TYPE OF ARTIFACT	ACCESSIBILITY TOOLS USED & VALIDATED BY ICUBE	ADDITIONAL RECOMMENDED TOOLS	ADDITIONAL COMMENTS
Microsoft Word	<u>Microsoft Accessibility Checker</u> ; <u>NVDA</u> ; <u>TPGi Color Contrast Analyzer (CCA)</u>	<u>JAWS</u> ; <u>AED CoP Baseline Guide/Checklist</u> ; <u>ICT Baseline Tests</u> ; <u>Manual Section 508 Review</u>	1. Microsoft Accessibility Checker is the primary accessibility validation tool for Microsoft Word documents. 2. NVDA and JAWS support screen reader testing. 3. Federal agencies commonly supplement automated testing with ICT Baseline Tests, AED CoP guidance, and manual Section 508 reviews to evaluate reading order, document structure, alternative text, and overall accessibility compliance. 4. TPGi CCA can be used to evaluate color contrast in diagrams, charts, and other visual elements.
PDF	<u>Adobe Acrobat Accessibility Checker</u> ; <u>PAC</u> ; <u>NVDA</u> ; <u>TPGi Color Contrast Analyzer (CCA)</u>	<u>JAWS</u> ; <u>CommonLook PDF Validator</u> ; <u>Manual Section 508 Review</u> ; <u>ICT Baseline Tests</u> ; <u>VoiceOver</u>	1. Adobe Acrobat Accessibility Checker and PAC are among the most widely used federal tools for PDF accessibility remediation and validation. 2. NVDA, JAWS, and VoiceOver are widely used for screen reader verification of PDF content across Windows and Apple platforms. 3. TPGi CCA can be used to evaluate color contrast in PDF content and visual elements.
Figma	<u>Stark</u> ; <u>TPGi Color Contrast Analyzer (CCA)</u> ; <u>Native Figma Accessibility Features</u> ; <u>NVDA</u> (interactive prototypes)	Browser Developer Tools; <u>axe for Designers (Deque)</u> ; <u>Accessibility Checker by UXMate</u> ; <u>ANDI</u> (after implementation); <u>JAWS</u> ; <u>VoiceOver</u>	1. Stark supports accessibility-focused design reviews, annotations, and contrast checks. 2. TPGi CCA is commonly used to validate color contrast for text and visual elements. 3. Native Figma accessibility features support review of contrast, components, labels, and focus order. 4. NVDA, JAWS, and VoiceOver are primarily used for testing interactive prototypes and implemented interfaces. 5. Browser Developer Tools, ANDI, axe for Designers, and Accessibility Checker by UXMate provide additional accessibility analysis after implementation. 6. Federal accessibility practices encourage use of USWDS, accessible component libraries, and accessibility validation throughout the design process. 7. Final Section 508 and WCAG conformance is determined on the implemented website or application—not within Figma itself.
Website / Web Application	<u>ANDI</u> ; <u>axe DevTools (Deque)</u> ; <u>WAVE</u> ; <u>NVDA</u> ; <u>TPGi Color Contrast Analyzer (CCA)</u>	<u>Google Lighthouse</u> ; <u>JAWS</u> ; <u>VoiceOver</u> ; <u>Accessibility Requirements Tool (ART)</u>	1. ANDI, axe DevTools, and WAVE are widely used for Section 508 and WCAG accessibility testing of websites and web applications. 2. TPGi CCA is commonly used to validate color contrast for text, controls, charts, icons, and other visual elements. 3. NVDA, JAWS, and VoiceOver are widely used for screen reader verification across Windows, MacOS, and mobile platforms. 4. Google Lighthouse provides automated accessibility analysis and identifies common accessibility issues. 5. ART helps identify applicable Section 508 requirements for ICT procurement and planning. 6. Accessibility testing includes keyboard navigation, focus order, headings, forms, tables, alternative text, error handling, & screen reader compatibility. 7. Federal accessibility validation combines automated testing, screen reader testing, and manual WCAG and Section 508 review.
Microsoft Forms	<u>Microsoft Accessibility Checker</u> (where applicable); <u>NVDA</u> ; <u>TPGi Color Contrast Analyzer (CCA)</u>	<u>ANDI</u> (Accessible Name & Description Inspector); <u>JAWS</u> ; <u>VoiceOver</u> ; <u>axe DevTools (Deque)</u> ; <u>WAVE</u>	1. Microsoft Accessibility Checker supports accessibility validation of related Microsoft 365 content where applicable. 2. TPGi CCA is commonly used to validate color contrast for text, controls, and visual elements. 3. NVDA, JAWS, and VoiceOver are widely used for screen reader verification and form usability testing. 4. ANDI, axe DevTools, and WAVE are commonly used to evaluate deployed web-based forms for WCAG and Section 508 compliance. 5. Accessibility testing includes labels, instructions, keyboard navigation, focus order, error handling, and screen reader compatibility. 6. Final accessibility compliance should be validated in the deployed browser experience.
Microsoft Excel	<u>Microsoft Accessibility Checker</u> ; <u>NVDA</u> ; <u>TPGi Color Contrast Analyzer (CCA)</u>	<u>JAWS</u> ; <u>ICT Baseline Tests</u> ; <u>VoiceOver</u> ; <u>Adobe Acrobat Pro Accessibility Checker</u> (for exported PDFs)	1. Microsoft Accessibility Checker is the primary accessibility validation tool for Excel workbooks. 2. TPGi CCA is commonly used to evaluate color contrast in charts, graphs, conditional formatting, and other visual elements. 3. NVDA, JAWS, and VoiceOver are widely used for screen reader verification of worksheets, tables, charts, and workbook navigation. 4. Accessibility testing includes table structure, worksheet names, headers, formulas, reading order, and alternative text. 5. ICT Baseline Tests provide standardized procedures for evaluating Section 508 compliance. 6. Excel files exported to PDF may be further validated using Adobe Acrobat Pro Accessibility Checker. 7. Federal accessibility validation combines automated testing, screen reader testing, and manual Section 508 review.